LiquidHaskell: Experience with Refinement Types in the Real World

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Haskell Background

Haskell's Type System

Haskell is a **statically typed** language.

- Every expression has a type.
- Types are checked before the program runs.
- Types are often inferred automatically.

Basic Types

Some common built-in types:

```
Int        -- integer
Float      -- floating point number
Double     -- floating point number
Num        -- [Int, Float, Double, ...]
Bool       -- True or False
Char       -- single character
String     -- text
```

Type Annotations

The operator `::` means “has type”.

```
age :: Int
age = 23

name :: String
name = "Mark"

active :: Bool
active = True
```

Function Types

The arrow (→) separates input and output types.

```
add :: Int -> Int -> Int
add x y = x + y
```

- First Int: first argument
- Second Int: second argument
- Last Int: result

Every function takes one argument: `Int → Int → Int` means `Int → (Int → Int)`, so `add 1` is a function `Int → Int`.

Roughly equivalent (uncurried) in a C-like language:

```
int add(int x, int y) { return x + y; }
```

Type Inference

Haskell can infer many types automatically.

```
1 age = 23
2 double x = x * 2
```

In GHCi we can inspect inferred types:

```
1 :t True
2 True :: Bool
3
4 :t "hello"
5 "hello" :: String
6
7 :t double
8 double :: Num a => a -> a
```

Limits of The Type System

We can’t define constraints for the types of a function like:

- Conditions about the value: `x > 0`
- Relationships between values: `a > b`

To check for positivity we need a dynamic test:

```haskell
sqrtRoot :: Double -> Double
sqrtRoot x
  | x > 0 = sqrt x
  | otherwise = error "x must be greater than 0"
```

Refinement Types and LiquidHaskell

LiquidHaskell

Many of these properties and relationships can be verified using a combination of **Refinement Types** and **external SMT solvers**.

This paper introduces and experiments with **LiquidHaskell** a refinement typechecker used to verify more than **10,000** lines of Haskell code across popular libraries.

Including: *bytestring*, *text*, *vector-algorithms* and *xmonad*.

Refinement Types

Refinement types:

- contain a **predicate** which is assumed to be true for any element of the refined type.
- can express **preconditions** when used as function arguments or **postconditions** when used as return types.

Examples:

positive natural numbers

$x : \{n \in \mathbb{N} \mid n > 0\}$

even natural numbers

$x : \{n \in \mathbb{N} \mid n \bmod 2 = 0\}$

result is a positive number

$f : \mathbb{N} \to \{y \in \mathbb{N} \mid y > 0\}$

result > input

$f(x : \mathbb{N}) : \mathbb{N} \to \{y \in \mathbb{N} \mid y > x\}$

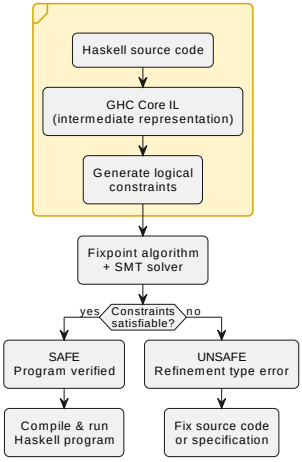
LiquidHaskell Architecture

Core

- Core is a simpler intermediate representation (GHC translates Haskell source to it)
- Source locations are preserved for error reporting

Constraints

- Refinement types are translated into logical constraints
- These constraints describe the properties that must hold



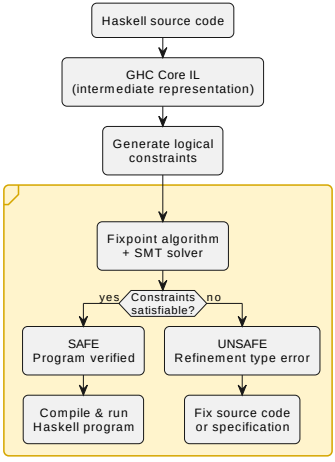
LiquidHaskell Architecture

Solution

- A fixpoint algorithm processes the constraints
- Any *SMT-LIB2* compatible solver checks whether the constraints can be satisfied (eg.: *Z3*, *CVC4*, …)

Result

- **SAFE** or **UNSAFE**: based on the verification of the refinements
- Errors are mapped back to their source-code locations



LiquidHaskell

LiquidHaskell is an **optional** type checker for Haskell. Refinement types are written inside **comments**, so they are optional and normal Haskell compilers can ignore them:

```
1 {-@ ... @-}
```

Execution

LiquidHaskell performs static verification before the Haskell program is executed. It does not become part of the executable or perform any checks during runtime.

LiquidHaskell Basics

```
1  {-@ type Natural = {v:Int | v >= 0} @-}
2  {-@ type Pos = {v:Int | v > 0} @-}
3
4  {-@ increment :: Natural -> Pos @-}
5  increment :: Int -> Int
6  increment x = x + 1
7
8  good1 = increment 0
9  good2 = increment 4
10
11 -- verification will fail here stating: Type Mismatch
12 bad = increment (-1)
```

LiquidHaskell Basics

```
1  {-@ type Natural = {v:Int | v >= 0} @-}
2  {-@ type Pos = {v:Int | v > 0} @-}
3
4  {-@ divide :: n:Natural -> d:Pos -> {v:Natural | v <= n} @-}
5  divide :: Int -> Int -> Int
6  divide x y = x `div` y
7
8  good = divide 5 2
9
10 -- With LH -> Type Mismatch before executing the code
11 -- Without LH -> runtime error: divide by zero
12 bad = divide 5 0
```

LiquidHaskell Parameterized Types

Type parameters are written as lowercase and **value parameters** are written as uppercase.

- **Type parameter:** *Int*, *Bool*, *[Int]*
- **Value parameter:** *0*, *10*, *100*

```
1 {-@ type NonEmpty a = {v:[a] | len v > 0} @-}
2
3 {-@ numbers :: NonEmpty Int @-}
4 numbers :: [Int]
5 numbers = []
6
7 {-@ letters :: NonEmpty Char @-}
8 letters :: [Char]
9 letters = ['a', 'b', 'c']
```

LiquidHaskell Parameterized Types

```
1  {-@ type Rng Low High = {v:Int | v >= Low && v <= High} @-}
2
3  {-@ grade :: Rng 0 100 -> Int @-}
4  grade :: Int -> Int
5  grade score
6      | score >= 90 = 5
7      | score >= 75 = 4
8      | score >= 60 = 3
9      | score >= 50 = 2
10     | otherwise   = 1
11
12 good = grade 85
13 bad = grade 120
```

LiquidHaskell Type Aliases

Type Aliases: used for reusability and readability.

```
1  -- with type alias
2  {-@ predicate Between Low High V = V >= Low && V <= High @-}
3  {-@ type Rng Low High = {v:Int | Between Low High v} @-}
4
5  -- we can use the predicate inline as well
6  {-@ grade :: Rng 0 100 -> {v:Int | Between 1 2 v} @-}
7  grade :: Int -> Int
8  grade score
9      | score >= 50 = 2
10     | otherwise   = 1
11
12 good = grade 85
13 bad = grade 120
```

Refining Data and Interfaces

LiquidHaskell Measures

So far, the specifications have been limited to comparisons and arithmetic operations on **primitive** values. **Measures** describe **inductive properties** of algebraic data types.

```
1 {-@ measure myLen @-}
2 myLen :: [a] -> Int
3 myLen []     = 0
4 myLen (x:xs) = 1 + myLen xs
```

- One equation for each data constructor: `[]` and `(:)`.
- Each right-hand side must belong to the restricted refinement logic.
- `measure myLen` makes this Haskell function available in refinements.

LiquidHaskell Measures: Minimum Length

```
1 {-@ type MinLength a N = {v:[a] | myLen v >= N} @-}
2
3 {-@ good :: MinLength Int 3 @-}
4 good :: [Int]
5 good = [1, 2, 3, 4]
6
7 {-@ bad :: MinLength Int 3 @-}
8 bad :: [Int]
9 bad = [1, 2]
```

- `a` is the element type; `N` is the minimum length.
- **SAFE:** `good` has length 4, and $4 \geq 3$.
- **UNSAFE:** `bad` has length 2, and $2 < 3$.

LiquidHaskell Measures: Constructor Refinements

Each list constructor gives LiquidHaskell a **length fact**:

```
[]  :: {v:[a] | myLen v == 0}

(:) :: x:a -> xs:[a]
    -> {v:[a] | myLen v == 1 + myLen xs}
```

- `[]`: the length is zero.
- `x:xs`: the length is one more than `xs`.
- These facts are available when building or pattern matching on lists.

LiquidHaskell Measures: Using a Constructor Fact

Adding an element increases the minimum length.

```
1 {-@ prepend :: a -> MinLength a 2 -> MinLength a 3 @-}
2 prepend :: a -> [a] -> [a]
3 prepend x xs = x : xs
4
5 good = prepend 1 [2, 3]
6 bad = prepend 1 [2]
```

- Input contract: $myLen\ xs \geq 2$.
- Constructor fact: $myLen\ (x : xs) = 1 + myLen\ xs \geq 3$.

LiquidHaskell Refined Data Types

Every value of a data type may need to satisfy an **invariant**.

```
1 data CSV a = CSV { cols :: [String], rows :: [[a]] }
```

- `cols`: the column names in the header.
- `rows`: the table's rows, containing values of type `a`.
- Haskell allows rows of different lengths.

CSV invariant

Every row must have exactly as many entries as the header has column names.

LiquidHaskell Refined Data Types: Definition

A field's refinement can refer to an **earlier field**.

```
1 {-@ type ListL a X = {v:[a] | len v == len X} @-}
2
3 {-@ data CSV a = CSV { cols :: [String]
4                     , rows :: [ListL a cols] } @-}
```

- `ListL a X`: a list with the same length as `X`.
- `rows`: a list of rows, each matching the length of `cols`.
- `len` is LiquidHaskell's built-in list length measure.

LiquidHaskell Refined Data Types: Verification

```
1 good = CSV ["Month", "Days"]
2              [ ["Jan", "31"]
3              , ["Feb", "28"]
4              , ["Mar", "31"] ]
5
6 bad =  CSV ["Month", "Days"]
7             [ ["Jan", "31"]
8             , ["Feb", "28"]
9             , ["Mar"] ]
```

LiquidHaskell Refined Type Classes

A **type class** defines a shared interface for different types.

```
1 class Reward a where
2   reward :: a -> Int -> Int
```

- `a`: the type that determines the reward policy.
- `reward`: takes a policy value and the current points.
- Each instance provides its own implementation.

Shared rule: points must be non-negative and cannot decrease.

LiquidHaskell Refined Type Classes: Contract

Refinements specify a **shared contract** for the class method.

```
{-@ class Reward a where
      reward :: a -> points:{Int | points >= 0}
             -> {v:Int | v >= points}
@-}
```

- **Precondition:** input points must be non-negative.
- **Postcondition:** the result cannot be smaller than the input.
- Every instance must satisfy the **same contract**.

LiquidHaskell Refined Type Classes: Bool Instance

A Boolean flag decides whether to add a bonus.

```
1 instance Reward Bool where
2   reward bonus points
3     | bonus     = points + 10
4     | otherwise = points
```

- `True`: add 10 points.
- `False`: keep the original points.
- Both branches satisfy $v \geq \mathit{points}$.

LiquidHaskell checks the instance against the class contract.

LiquidHaskell Refined Type Classes: Verification

```
1 good :: Int
2 good = reward True 20
3
4 bad :: Int
5 bad = reward True (-1)
```

- **SAFE:** `good` passes non-negative points and returns 30.
- **UNSAFE:** `bad` violates the precondition $points \geq 0$.

LiquidHaskell Refined Type Classes: Invalid Instance

What if we replace the bonus with a deduction?

```haskell
-- alternative implementation:
instance Reward Bool where
  reward bonus points
    | bonus     = points - 10
    | otherwise = points
```

- The first branch violates $v \geq points$.
- Callers must meet the precondition.
- Instances must prove the postcondition.

Totality and Advanced Termination

LiquidHaskell Termination

Haskell only evaluates expressions when their results are needed (*non-strict evaluation*).

- Some functions may run forever without producing a result.
- **LiquidHaskell:** Must ensure that functions terminate to safely verify certain properties.
- By default, LiquidHaskell proves termination of each recursive function.

Termination metric

A termination metric is a value that:

- Gets smaller with **every** recursive call.
- Cannot decrease forever (e.g., a non-negative integer).

LiquidHaskell Termination: Factorial

A simple termination metric is a **non-negative integer**.

```
1 {-@ fac :: n:Nat -> Nat / [n] @-}
2 fac :: Int -> Int
3 fac 0 = 1
4 fac n = n * fac (n - 1)
```

- `Nat`: integers greater than or equal to zero.
- `/ [n]`: use the argument `n` as the termination metric.
- The base case returns immediately.
- The recursive case decreases `n` by 1 each call.

LiquidHaskell Termination: Processing Batches

Consider a simplified **batch worker** that counts completed jobs.

- `b`: batches remaining, including the current batch.
- `j`: jobs remaining in the current batch.
- Completing a job decreases `j`, while `b` stays unchanged.
- Moving to the next batch decreases `b` and resets `j` to 3.

Two batches of three jobs

$(2,3) \to (2,2) \to (2,1) \to (2,0) \to (1,3) \to \cdots \to (1,0)$.

Neither argument strictly decreases at every call. We supply a **custom termination metric**: the pair `[b, j]`.

LiquidHaskell Termination: Lexicographic Ordering

Compare the **batch count first**, then the job count.

```
{-@ jobs :: b:Nat -> j:Nat -> Nat / [b, j] @-}
jobs :: Int -> Int -> Int
jobs b j
  | b == 0    = 0
  | j > 0     = 1 + jobs b (j - 1)
  | b == 1    = 0
  | otherwise = jobs (b - 1) 3
-- jobs 2 3 == 6
```

LiquidHaskell Termination: Why the Pair Decreases

`/ [b, j]` uses **lexicographic ordering**: compare `b` first, and compare `j` only when `b` is unchanged.

- `jobs b (j - 1)`: the first component stays equal, the second decreases.
- `jobs (b - 1) 3`: the first component decreases.
- Both components remain non-negative.
- At `b == 0`, or when the last batch is empty (`b == 1` and `j == 0`), recursion stops.

Functional Correctness Invariants

LiquidHaskell Functional Correctness

Beyond avoiding errors, refinements can express **application-specific correctness rules**.

The paper's example: **red-black trees**, a balanced binary search tree structure.

```
1 data Col    = R | B
2 data Tree a = Leaf
3             | Node Col a (Tree a) (Tree a)
```

- `Col`: the color; `a`: the key; followed by the left and right subtrees.

LiquidHaskell Red-Black Trees: Three Invariants

A tree is only a valid Red-Black tree if:

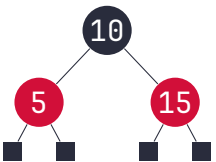
- **Order:** all keys in the left subtree are smaller than the node's keyand all keys in the right subtree are larger.
- **Color:** a red node has only black children. Empty leaves count as black.
- **Height:** from any node, every path to a leaf contains the same number of black nodes.

Main idea

Describe these properties separately, then combine them without changing the underlying tree structure.

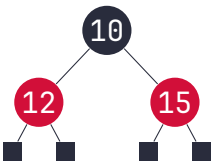
LiquidHaskell Red-Black Trees: Valid and Invalid

Valid



$5 < 10 < 15$.
Color and height are valid too.

Invalid



Order fails: $12 > 10$.

Black squares represent empty leaves.

LiquidHaskell Red-Black Trees: Checking Color

```
1 measure isB :: Tree a -> Prop
2 isB (Node c x l r) = c == B
3 isB (Leaf) = true
4
5 measure isRB :: Tree a -> Prop
6 isRB (Leaf) = true
7 isRB (Node c x l r) = isRB l && isRB r &&
8                       c == R => (isB l && isB r)
```

- `isB`: the root is black; empty leaves count as black.
- `isRB`: red nodes have black children, recursively.

LiquidHaskell Red-Black Trees: Checking Height

```
1 measure bh :: Tree a -> Int
2 bh (Leaf) = 0
3 bh (Node c x l r) = bh l + if c == R then 0 else 1
4
5 measure isBal :: Tree a -> Prop
6 isBal (Leaf) = true
7 isBal (Node c x l r) = bh l == bh r && isBal l && isBal r
```

- `bh`: count black nodes along the leftmost path.
- `isBal`: require equal black heights, recursively.

LiquidHaskell Red-Black Trees: Checking Order

```
1  data Tree a <l :: a -> a -> Prop,
2              r :: a -> a -> Prop>
3    = Leaf
4    | Node { c   :: Col
5           , key :: a
6           , lt  :: Tree <l, r> a <l key>
7           , rt  :: Tree <l, r> a <r key> }
8  type OTree a
9    = Tree <{\k v -> v < k},   -- l: left rule
10          {\k v -> v > k}> a -- r: right rule
```

- `l` and `r`: rules relating keys to subtree values.
- `OTree`: left keys $v < k$; right keys $v > k$.

LiquidHaskell Red-Black Trees: Combining the Rules

Combining ordering, color, and black-height balance:

```
1 type RBT a = {v:OTree a | isRB v && isBal v}
```

- `OTree`: keys follow binary-search ordering.
- `isRB`: red nodes have black children.
- `isBal`: all paths have equal black heights.

With these invariants and suitable function signatures, LiquidHaskell verifies the library's **insertion, deletion, and rebalancing** procedures.

Evaluation

LiquidHaskell Evaluation: Benchmarks

Library	Verified properties
GHC.List, Data.List	size properties of list operations
Data.Set.Splay	termination, well-ordered splay trees
Data.Map.Base	termination, binary-search ordered trees
HsColour	totality of all functions
XMonad	window uniqueness, some QuickCheck properties
ByteString	termination, memory safety, correctness
Text	pointer safety, correctness (found a bug)
Vector-Algorithms	safe vector accessing, indexing and slicing

LiquidHaskell Evaluation: Results

Module	LOC	Mod	Fun	Specs	Annot	Qualif	Time (s)
GHC.List	309	1	66	29 / 38	6 / 6	0 / 0	15
Data.List	504	1	97	15 / 26	6 / 6	3 / 3	11
Data.Map.Base	1396	1	180	125 / 173	13 / 13	0 / 0	174
Data.Set.Splay	149	1	35	27 / 37	5 / 5	0 / 0	27
HsColour	1047	16	234	19 / 40	5 / 5	1 / 1	196
XMonad.StackSet	256	1	106	74 / 213	3 / 3	4 / 4	27
ByteString	3505	8	569	307 / 465	55 / 55	47 / 124	294
Text	3128	17	493	305 / 717	52 / 54	49 / 97	499
Vector-Algorithms	1218	10	99	76 / 266	9 / 9	13 / 13	89
Total	11512	56	1879	977 / 1975	154 / 156	117 / 242	1336

Specs, Annot, Qualif: count / lines of type specifications, other annotations (invariants, termination hints) and qualifiers.

LiquidHaskell Evaluation: What the Numbers Show

- **11,512** lines of code in 56 modules, checked with **1,975** lines of specifications.
- Hints (annotations and qualifiers) add only **3.5%** of the code size.

Performance

The full suite takes about **22 minutes**, but checking a handful of functions in isolation is fast enough for **interactive use**.

LiquidHaskell Limitations

The case studies also showed limitations of LiquidHaskell.

- Ghost parameters
- Fixed-width integers and floating-point numbers
- Higher-order functions
- Functions as data
- Lazy binders
- Assumes
- Error messages

Workaround

In most cases, the code could be **altered slightly** to make verification possible.

References

- Niki Vazou, Eric L. Seidel, Ranjit Jhala. *LiquidHaskell: Experience with Refinement Types in the Real World*. Haskell Symposium, 2014.
- LiquidHaskell documentation: https://ucsd-progsys.github.io/liquidhaskell